
3AL Hand-in 4 2024 Solutions

Due: Friday 3 May 11:00 pm

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false. No explanation needed.

- (a) If G is a finite group and $p \mid |G|$, then all elements of order p are contained in a Sylow p -subgroup.
- (b) If H is a Sylow p -subgroup of a group with $|H| = p^n$, then all elements with order p^n are contained in H .
- (c) The number of conjugates of a Sylow p -subgroup is a multiple of p .
- (d) If G has an abelian Sylow p -subgroup, then all Sylow p -subgroups of G are abelian.

(a) True. All elements of order p are contained in a p -subgroup (consider the cyclic subgroup generated by that element), all p -subgroups are contained in a Sylow p -subgroup.

(b) False. e.g. D_3 has a Sylow 2-subgroup $\{1, s\}$ but sr is an element of order 2 not contained in this Sylow 2-subgroup.

(c) False. By the Second Sylow Theorem, the number of conjugates of a Sylow p -subgroup is the number of Sylow p -subgroups. By the Third Sylow Theorem, $n_p \equiv 1 \pmod{p}$, thus $p \nmid n_p$.

(d) True. Any conjugate of an abelian subgroup is also abelian (since inner automorphisms are isomorphisms).

[4 marks]

2. Use the Sylow Theorems to prove that if G is a group of order 33, then $G \cong \mathbb{Z}_3 \times \mathbb{Z}_{11}$.

Let G be a group of order $33 = 3 \times 11$. By the First Sylow Theorem, G has Sylow 3- and 11-subgroups. By the Third Sylow Theorem, $n_3 \equiv 1 \pmod{3}$ and $n_3 \mid 11$. Thus

$n_3 = 1$. So G has a unique Sylow 3-subgroup, P . By the Second Sylow Theorem, $P \triangleleft G$. Similarly, $n_{11} \equiv 1 \pmod{11}$ and $n_{11} \mid 3$, thus $n_{11} = 1$ and so there is a unique Sylow 11-subgroup $Q \triangleleft G$.

By Lagrange's Theorem, we have $|P \cap Q|$ must divide 3 and 11. Thus $|P \cap Q| = 1$. Thus $|PQ| = |P||Q|/|P \cap Q| = 33$ and so $PQ = G$. Thus $G \cong P \times Q$. Since P and Q both have prime order, they are cyclic and isomorphic to $\mathbb{Z}_3$ and $\mathbb{Z}_{11}$, respectively. Thus $G \cong \mathbb{Z}_3 \times \mathbb{Z}_{11}$.

[8 marks]

Maximum mark = 10

Available marks = 12